

Time-Reversal Symmetry in Transformer Predictions on the Cylinder Graph HMM

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Contents

1	Introduction	2
2	Theoretical Background	3
2.1	Entropy Rate Invariance	3
2.2	Reverse HMM Construction	3
2.3	Belief-State Convergence and Finite-Context Effects	3
3	Hypotheses	4
4	Experimental Setup	4
4.1	HMM and Dataset	4
4.2	Architecture Sweep	5
4.3	Matched Comparison	5
5	Theoretical Predictions	5
6	Results	6
6.1	H3: Is the Reversed Direction Harder?	6
6.2	H4: Architecture Dependence of the Gap	8
6.3	Architectural Breakdown	8
6.4	Interpretation	9
7	Activation Ablation Experiments	9
7.1	Overview	9
7.2	Direction Identification	9
7.3	Ablation Protocol	10
7.4	Results	10
8	Embedding Geometry	12
8.1	Cosine Similarity Structure	12
8.2	Norm Structure	12
8.3	Relationship to PMI Embeddings	12

9	Training Dynamics	14
9.1	Setup	14
9.2	H5: Transient Dynamics Asymmetry	14
9.3	H6: Learning Phases	14
9.4	H7: Architecture-Dependent Convergence Gap	17
9.5	Training Dynamics: Summary	17
9.6	Short-Context Follow-Up: $T = 4$	17
10	Discussion and Open Questions	22
10.1	Summary of Findings	22
10.2	Open Questions	23

Yes

1 Introduction

A natural question in the study of transformers trained on Hidden Markov Model (HMM) sequences is whether the *direction* of time affects learnability. Information-theoretically, the entropy rate of a stationary process is invariant under time reversal: the forward and backward sequences contain the same amount of uncertainty per token. A Bayes-optimal predictor with infinite context therefore achieves the same cross-entropy loss in both directions.

However, a finite-context transformer may find one direction systematically easier if the latent belief state converges to its stationary distribution at a different rate in each temporal direction. The convergence rate depends on the spectral properties of the transition matrix, which is generally different for the forward and reverse chains.

This report investigates that question on the Cylinder Graph HMM (fully described in `notes/model_comparison` whose directed transition structure creates an a priori asymmetry between the forward and reverse processes. We formulate seven testable hypotheses (Section 3), describe the experimental setup (Section 4), present theoretical predictions including the Bayes-optimal gap at context length $k = 16$ (Section 5), and compare against a matched architecture sweep (Section 6). We also report complementary experiments on belief-state localisation via activation ablations (Section 7), on the structure of the learned token-embedding geometry (Section 8), and on forward-reversed training dynamics (Section 9). **This report investigates that question on the Cylinder Graph HMM (fully described in `notes/model_comparison/cylinder_hmm_model_comparison.tex`), whose directed transition structure creates an a priori asymmetry between the forward and reverse processes. We formulate seven testable hypotheses (Section 3), describe the experimental setup (Section 4), present theoretical predictions including the Bayes-optimal gap at context length $k = 16$ (Section 5), and compare against a matched architecture sweep (Section 6). We also report complementary experiments on belief-state localisation via activation ablations (Section 7), on the structure of the learned token-embedding geometry (Section 8), on forward-reversed training dynamics (Section 9), and on a short-context follow-up at $T = 4$ designed to test whether the belief-state asymmetry affects optimisation in the regime where it is largest (Section 9.6).**

2 Theoretical Background

2.1 Entropy Rate Invariance

For a stationary stochastic process $\{X_t\}$, the entropy rate is

$$h = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, \dots, X_n). \quad (1)$$

Because the joint entropy $H(X_1, \dots, X_n)$ depends only on the joint distribution and not on the ordering of its arguments, the entropy rate of the forward process $(\dots, X_{t-1}, X_t, X_{t+1}, \dots)$ is identical to that of the time-reversed process $(\dots, X_{t+1}, X_t, X_{t-1}, \dots)$.

Consequence. Any difference in transformer loss between forward and backward prediction cannot be explained by an inherent information-theoretic asymmetry. The Bayes-optimal predictor with infinite context achieves identical loss in both directions.

2.2 Reverse HMM Construction

Given a forward HMM with joint emission–transition matrices $E[j, i, k] = P(x_t=j, s_{t+1}=k \mid s_t=i)$ and stationary distribution π , the reverse-time HMM has emission matrices

$$E_{\text{rev}}[j, k, i] = \frac{\pi(i) \cdot E[j, i, k]}{\pi(k)}. \quad (2)$$

This follows from Bayes’ rule applied to the time-reversed Markov chain. The reverse HMM satisfies:

- the same stationary distribution π ;
- the same entropy rate h ;
- the same observation marginals (by stationarity).

2.3 Belief-State Convergence and Finite-Context Effects

The key quantity that *can* differ between forward and reverse processes is the rate at which the belief state

$$\alpha_t(s) = P(s_t=s \mid x_{t-k}, \dots, x_{t-1}) \quad (3)$$

converges as a function of the context length k . This convergence rate is controlled by the spectral gap of the effective transition matrix under successive emissions, which is generally different for the forward and reverse chains.

For the Cylinder Graph HMM specifically:

- **Forward:** Within-ring transitions go clockwise (+1, +2 mod n); inter-layer transitions go forward ($l \rightarrow l+1 \bmod D$). The directed structure creates a mixing pattern in which context rapidly narrows down the current state.
- **Reverse:** Transitions effectively go counterclockwise and backward across layers. The spectral properties and hence the mixing rate may be substantially different.

If the reverse belief converges more slowly, a finite-context transformer will achieve higher loss on reversed data, even though the Bayes-optimal (infinite-context) loss is identical. Let $L^*(k)$ denote the Bayes-optimal cross-entropy at context length k . The relevant comparison is

$$\Delta_{\text{Bayes}}(k) = L_{\text{rev}}^*(k) - L_{\text{fwd}}^*(k), \quad (4)$$

which measures the intrinsic difficulty gap at finite context.

3 Hypotheses

- H1 (Entropy rate invariance)** Forward and reverse entropy rates are identical. *Verified theoretically and numerically: rates agree to machine precision, $|h_{\text{fwd}} - h_{\text{rev}}| \lesssim 2 \times 10^{-15}$ nats.*
- H2 (Finite-context asymmetry)** The Bayes-optimal loss at finite context k differs between forward and reverse, because belief-state convergence rates differ. Preliminary calculations (500 samples, $k \leq 8$) show the KL divergence from the converged belief is $\sim 2\text{--}3\times$ larger at each context position for the reverse process.
- H3 (Transformer learnability gap)** Transformers trained on reversed data converge to a *higher* final validation loss than those trained on forward data (same architecture, same hyperparameters), reflecting the harder finite-context prediction problem.
- H4 (Architecture dependence)** The forward–reverse gap $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ depends on model capacity: larger models may close the gap by implementing a longer effective context.
- H5 (Transient dynamics asymmetry)** Even if the final validation losses are identical, the reversed direction should exhibit slower early-training improvement and a larger integrated excess loss over training. The physical analogy is two systems relaxing to the same equilibrium from different initial conditions: the reversed process, which requires more context to form accurate beliefs, should provide noisier gradient signals early in training.
- H6 (Learning phases)** Both loss curves decompose into Phase I (unigram learning, direction-independent), Phase II (context learning, potentially direction-dependent), and Phase III (saturation). H6 predicts that Phase II onset is delayed and/or prolonged for the reversed direction.
- H7 (Architecture-dependent convergence gap)** Models with more layers (longer effective context) should show a *smaller* convergence speed gap between forward and reversed.

4 Experimental Setup

4.1 HMM and Dataset

All experiments use the Cylinder Graph HMM with the parameters in Table 1.

Parameter	Value
Width n	6
Depth D	3
Tokens per cluster K	16
Vocabulary size $ V $	$D \times K = 48$
Dirichlet concentration α	0.3
Inter-layer probability p	0.1
Dataset size	10^7 tokens
Sequence length T	16

Table 1: Cylinder Graph HMM parameters used throughout.

The forward dataset is at `data/datasets/cylinder_graph_hmm/`. The reversed dataset (`data/datasets/cylinder_graph_hmm_reversed/`) is constructed by flipping each sequence in time: if the forward sequence is $(x_1, x_2, \dots, x_T)$, the reversed sequence is $(x_T, x_{T-1}, \dots, x_1)$.

4.2 Architecture Sweep

Both forward and reverse sweeps cover the same 36-run grid:

Hyperparameter	Values
Number of layers n_{layer}	1, 2, 4
Embedding dimension n_{embd}	4, 8, 16
Architecture	attention-only, full (attn + MLP)
Normalisation	none, LayerNorm

All models use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01, learning rate 10^{-3}), batch size 128, and 10 training epochs. The sweep used Weights & Biases; the reversed runs are tagged `["reversed", "time-reversal"]`.

4.3 Matched Comparison

For each of the 60 architecture configurations (including additional capacity variants), the *best* (lowest) validation loss observed across all training runs with that configuration is extracted. This yields 60 matched pairs $(L_{\text{fwd}}, L_{\text{rev}})$.

5 Theoretical Predictions

The script `src/reverse_hmm_analysis.py` constructs the reverse HMM via Eq. (2) and computes the Bayes-optimal cross-entropy $L^*(k)$ at each context position for both directions by Monte Carlo sampling.

Entropy rates. The joint entropy rate $H(X_t, S_{t+1} | S_t)$, computed analytically from the emission matrices, agrees to machine precision between forward and reverse ($|h_{\text{fwd}} - h_{\text{rev}}| \sim 2 \times 10^{-15}$ nats), confirming H1. The observation entropy rate h_X is identical for both directions by the permutation-invariance argument, and the empirical estimates ($h_X \approx 2.44$ nats) are consistent.

Belief-state convergence. The excess Bayes-optimal loss $L^*(k) - h_X$ quantifies how far the optimal finite-context predictor is from the infinite-context limit. Notably, the forward and reverse excess losses are nearly identical at all context lengths: both start at ~ 1.22 nats at $k = 1$ and decay to the Monte Carlo noise floor ($\lesssim 5 \times 10^{-3}$ nats) by $k \approx 6$. This stands in contrast to the belief-state KL divergence, which shows a persistent $\sim 2\times$ asymmetry (reverse beliefs converge more slowly). The decoupling suggests that the hidden states the reverse process is confused between have sufficiently similar emission distributions that the additional state uncertainty does not affect prediction quality.

Bayes-optimal gap at $k = 16$. A full Monte Carlo run with 50,000 samples and $k \leq 16$ yields:

Quantity	Value
$L_{\text{fwd}}^*(k=16)$	2.4362 nats
$L_{\text{rev}}^*(k=16)$	2.4368 nats
$\Delta_{\text{Bayes}}(k=16)$	+0.0006 nats

The Bayes-optimal gap has essentially vanished by $k = 16$: both directions are within 0.001 nats of each other at the training context length. This is a key result—it means the near-zero transformer loss gap (Section 6) is *fully explained* by the information-theoretic gap closing at this context length. The belief-state convergence asymmetry (H2) is real but irrelevant at $k = 16$: both forward and reverse Bayes-optimal predictors have converged to within ~ 0.002 nats of the observation entropy rate $h_X \approx 2.44$.

Note on the entropy rate. The “theoretical” entropy rate $-\sum_i \pi(i) \sum_{j,k} E[j, i, k] \log E[j, i, k] = 2.534$ nats is the entropy rate of the *joint* (X_t, S_{t+1}) process, i.e. $H(X_t, S_{t+1} | S_t)$. The observation entropy rate $h_X = H(X_t | X_{t-1}, X_{t-2}, \dots)$ is lower because observing past emissions gives only partial information about the current hidden state. We estimate $h_X \approx 2.44$ nats empirically via the forward algorithm on a sequence of 10^5 tokens.

Figure 1 summarises these theoretical results.

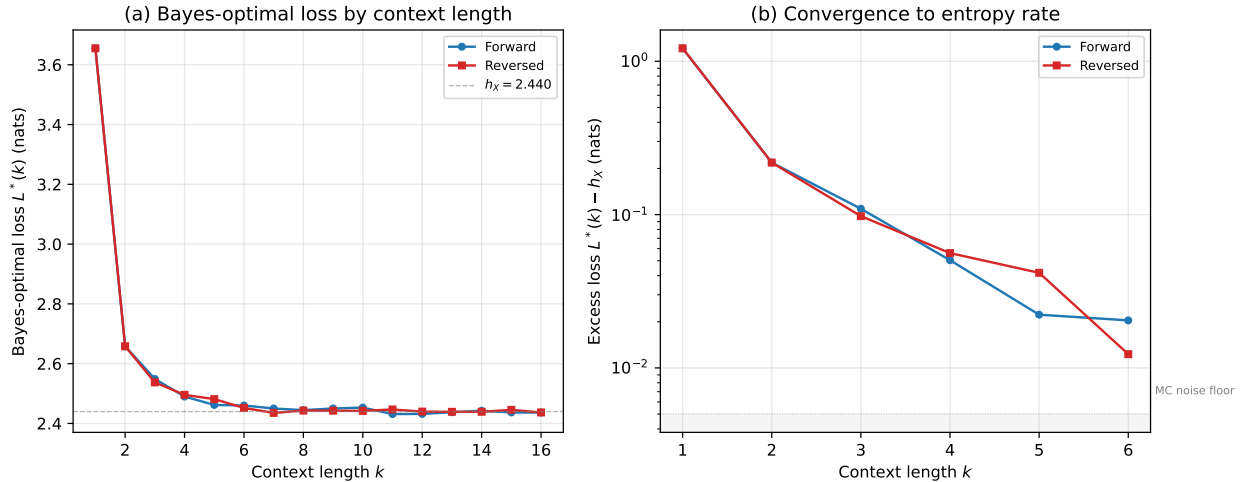
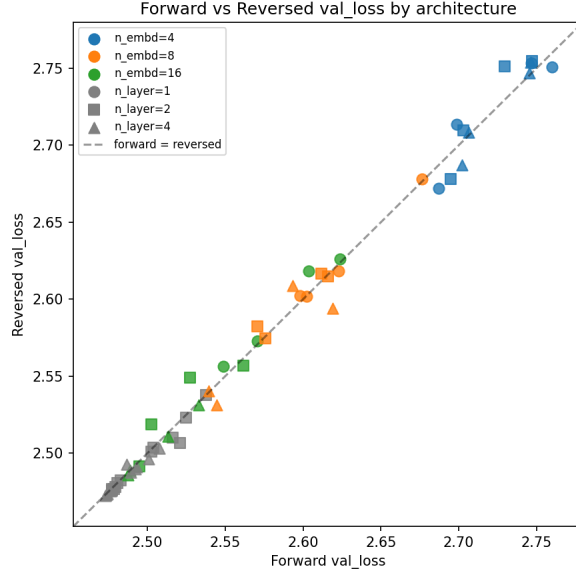


Figure 1: Bayes-optimal analysis of forward vs. reversed CylinderGraph HMM (50,000 Monte Carlo samples). (a) Bayes-optimal loss $L^*(k)$ converges to the observation entropy rate $h_X \approx 2.44$ nats for both directions by $k \approx 6$. (b) Excess Bayes-optimal loss $L^*(k) - h_X$ on a log scale. Forward and reverse excess losses are nearly identical at all k (both ~ 1.22 nats at $k = 1$), despite a $\sim 2\times$ asymmetry in belief-state KL convergence. Both reach the Monte Carlo noise floor ($\lesssim 5 \times 10^{-3}$ nats) by $k \approx 6$.

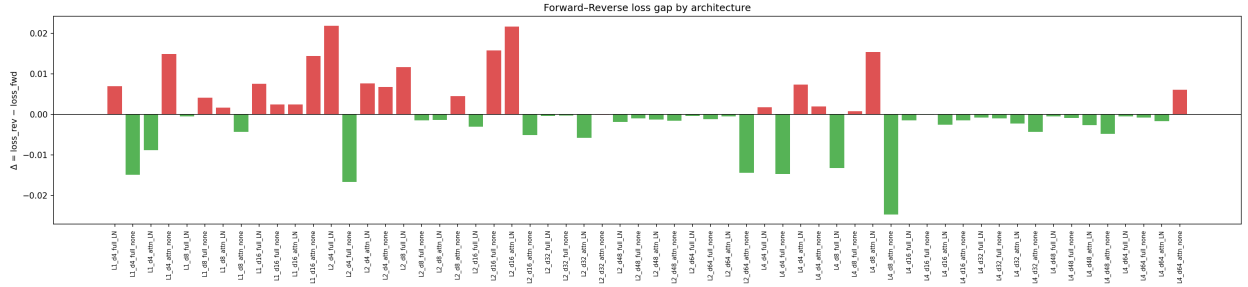
6 Results

6.1 H3: Is the Reversed Direction Harder?

Figure 2 shows the forward versus reversed validation loss for the 60 matched configurations. Table 2 summarises the statistics of $\Delta = L_{\text{rev}} - L_{\text{fwd}}$.



(a) Forward vs. reversed validation loss for all 60 matched configurations, coloured by n_{embd} . Points on the diagonal indicate identical performance; points above the diagonal indicate reversed sequences are harder. The data cluster tightly around the diagonal.



(b) $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ for each of the 60 configurations, sorted by magnitude. Most values cluster within ± 0.01 nats of zero. The extreme outlier ($\Delta \approx -0.025$) corresponds to the attention-only model L4_d8_attn_noLN.

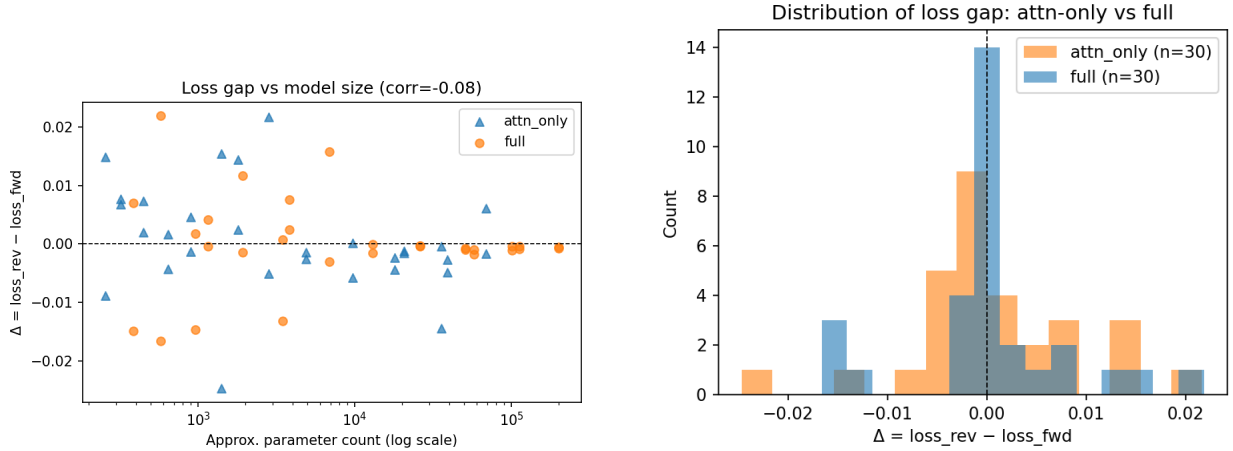
Figure 2: Forward versus reversed validation loss across the full architecture sweep. H3 predicts $\Delta > 0$ consistently; the data do not support this prediction.

Statistic	Value
Fraction $\Delta > 0$	22/60 (36.7%)
Mean $\pm$ s.d. of Δ	$+0.0002 \pm 0.0086$ nats
Minimum Δ	-0.025 nats
Maximum Δ	$+0.022$ nats

Table 2: Summary statistics of $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ across 60 matched configurations. The mean is consistent with zero.

Result: H3 is not supported. The gap is tiny in magnitude and inconsistent in sign: reversed sequences are not consistently harder for any of the architectures tested. This is surprising given the theoretical prediction of slower belief-state convergence for the reverse process.

6.2 H4: Architecture Dependence of the Gap



(a) Δ versus approximate parameter count. A slight downward trend indicates that larger models tend toward $\Delta \leq 0$, opposite to the prediction that larger models should *close* the gap.

(b) Distribution of Δ split by architecture type (attention-only vs. full transformer with MLP layers). Both distributions are centred near zero.

Figure 3: Dependence of the forward–reverse gap Δ on model capacity and architecture type.

Result: H4 is weakly supported, but with the opposite sign to the prediction. Table 3 shows the Pearson correlation of Δ with several capacity measures.

Capacity measure	Pearson r with Δ
n_{params} (approx.)	-0.083
n_{embd}	-0.148
n_{layer}	-0.205

Table 3: Correlation of Δ with model capacity measures. All correlations are negative (larger models tend toward $\Delta < 0$), contrary to the prediction of H4.

All correlations are negative: larger models marginally *favour* reversed sequences. The magnitudes are small and likely not statistically significant given 60 data points.

6.3 Architectural Breakdown

Figure 4 shows the mean validation loss by n_{layer} and n_{embd} for both conditions.

Notable patterns in the per-configuration breakdown:

- **Extreme outlier:** L4_d8_attn_only_noLN achieves $\Delta = -0.025$ nats (reversed is *easier*). This is the largest deviation in either direction.

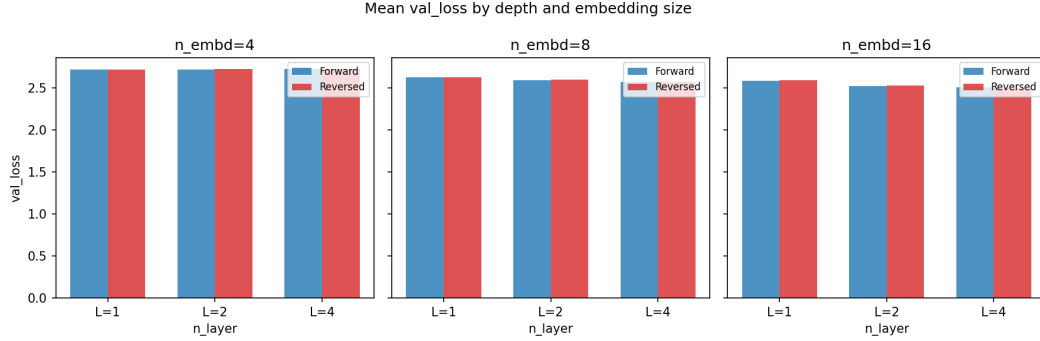


Figure 4: Mean validation loss grouped by n_{layer} and n_{embd} , for forward (blue) and reversed (orange) conditions. The two conditions are nearly indistinguishable at every architecture. The largest separations occur at small embedding dimension ($n_{\text{embd}} = 4$).

- **Small models** ($n_{\text{embd}} = 4$) tend toward positive Δ , consistent with H3.
- **Large full models** ($n_{\text{embd}} \geq 32$, full transformer with MLP layers) mostly have $\Delta \leq 0$.

6.4 Interpretation

The near-zero mean gap is now fully explained by the Bayes-optimal analysis (Section 5): $\Delta_{\text{Bayes}}(16) = +0.0006$ nats, meaning the information-theoretic gap has already closed at the training context length. **Interpretation 1 (context window is sufficient) is confirmed.** The belief-state convergence asymmetry is real (the reverse process needs $\sim 2\times$ more context) but irrelevant at $T = 16$ because both directions have converged. Since there is no Bayes-optimal gap to exploit, no transformer—regardless of architecture—can show a systematic forward–reverse difference.

7 Activation Ablation Experiments

7.1 Overview

A complementary set of experiments investigated whether the belief-state geometry learned by the transformer is spatially localised within the residual stream. Using the best trained model ($L = 4$, $d = 8$, $H = 1$, no LayerNorm), we identified the low-dimensional subspace of the residual stream most aligned with the mixed-state (belief-state) representation via PCA regression on the activations at the final sequence position.

7.2 Direction Identification

The “fractal directions” are the principal components of the residual-stream activations that best predict the mixed-state representation (the belief distribution over hidden states) at the final token position. The orthogonal complement forms the “non-fractal” subspace. The leading two principal components account for the bulk of the explained variance, consistent with the low-dimensional fractal geometry of the HMM belief simplex.

7.3 Ablation Protocol

At inference time the activation at one or more sequence positions is replaced by its projection onto the *orthogonal complement* of the fractal subspace—i.e. the fractal-direction component is zeroed out. The model is then run to completion and the resulting cross-entropy loss is recorded. Two variants were tested:

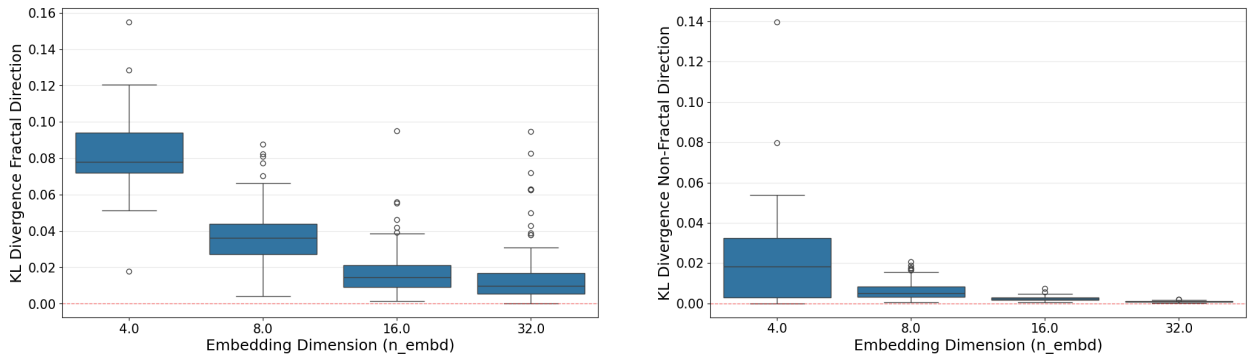
1. **Last-position ablation:** Only the activation at the final sequence position is intervened upon, consistent with conventional activation-patching practice.
2. **Full-sequence ablation:** Every position’s activation is replaced, preventing any accumulation of belief-state information along the entire sequence.

7.4 Results

The key finding is that **the fractal directions are substantially more important than the orthogonal subspace**:

- Ablating the fractal-direction component (zeroing the informative subspace) causes a much larger increase in cross-entropy than ablating the orthogonal complement. This asymmetry holds for both the last-position and full-sequence modes.
- Nevertheless, ablating the fractal direction does not entirely eliminate predictive performance, particularly for models with larger embedding dimensions. This suggests that larger models distribute information across a wider subspace.

Figures 5 and 6 show the last-position ablation results; Figure 7 shows the full-sequence ablation.



(a) Validation loss after zeroing the fractal-direction component at the last position, compared with zeroing the orthogonal complement (model variant A).

(b) Same as (a) for model variant B. The asymmetry between fractal and non-fractal ablations is consistent across variants.

Figure 5: Last-position ablation: cross-entropy loss when zeroing the fractal-direction or orthogonal-complement component of the residual stream at the final sequence position. Ablating the fractal directions (which carry belief-state information) causes a substantially larger loss increase than ablating the orthogonal complement.

These results confirm that the transformer compresses its belief-state representation into a low-dimensional subspace of the residual stream, consistent with the *mixed-state presentation* picture of HMM belief states.

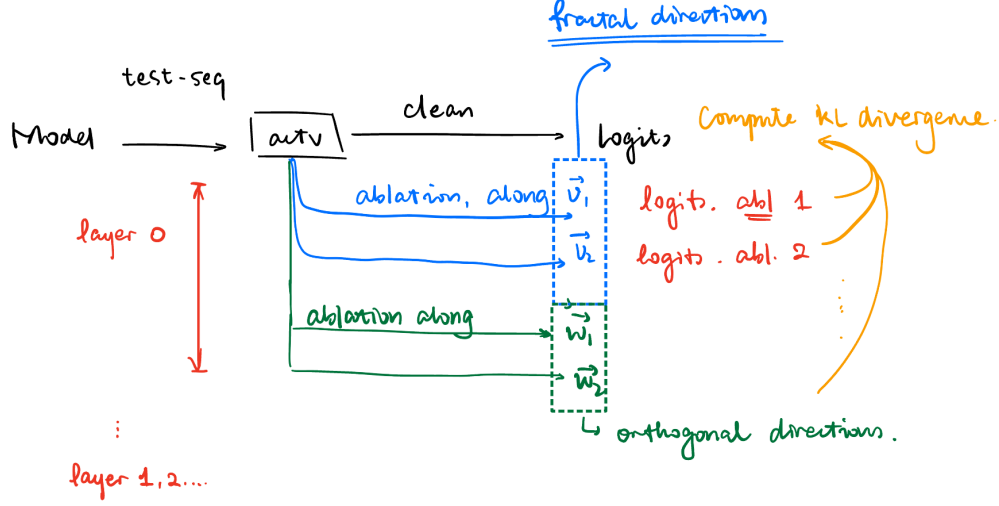
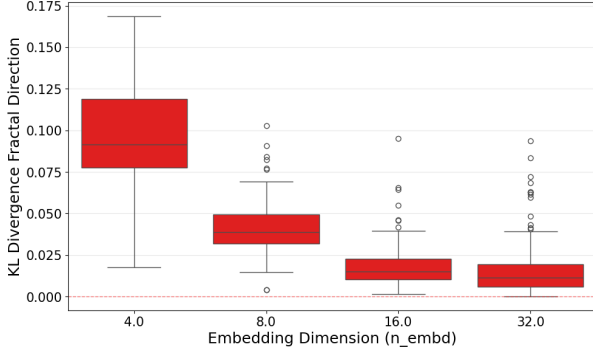
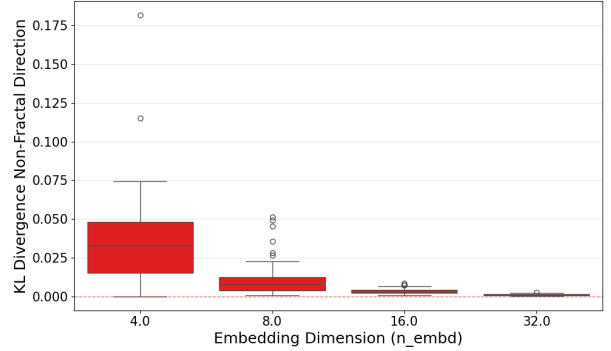


Figure 6: Summary of last-position ablation results across model sizes and configurations. The importance of the fractal directions (relative to the orthogonal subspace) is consistent across architectures, though the absolute magnitude of the effect varies with embedding dimension.



(a) Full-sequence ablation for model variant A: every position’s fractal-direction component is zeroed, preventing any accumulation of belief-state information.



(b) Full-sequence ablation for model variant B. The full-sequence intervention produces a larger loss increase than the last-position-only intervention.

Figure 7: Full-sequence ablation: the fractal-direction component is zeroed at *every* sequence position, fully blocking belief-state accumulation throughout the context window. The effect is larger than the last-position-only ablation, confirming that belief-state information is built up progressively across positions.

8 Embedding Geometry

8.1 Cosine Similarity Structure

Analysis of the learned token-embedding matrix $W_E \in \mathbb{R}^{48 \times d}$ reveals a strong 3×3 block structure in the cosine-similarity matrix: tokens within the same depth cluster are mapped to similar embedding directions, while cross-cluster pairs are near-orthogonal or anti-correlated. This is described in detail in `notes/model_comparison/cylinder_hmm_model_comparison.tex`; we summarise the key additional findings below.

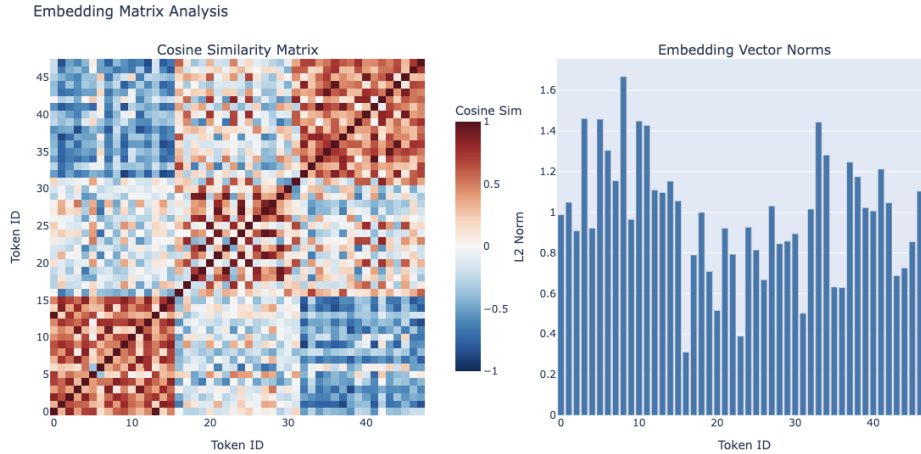


Figure 8: Cosine-similarity matrix of the learned token embeddings W_E . The 48 tokens are ordered by depth cluster (clusters of 16), revealing a clear 3×3 block structure: within-cluster cosine similarities are substantially higher than cross-cluster similarities. This mirrors the depth-level organisation of the Cylinder Graph HMM.

8.2 Norm Structure

The ℓ_2 norms of individual token embeddings do not correlate with the empirical token frequencies in the training data. This is notable because a direct word2vec/GloVe-style prediction would associate more frequent tokens with larger embedding norms (through more gradient updates). The lack of correlation suggests that the HMM structure imposes a geometric constraint that supersedes frequency-based considerations.

8.3 Relationship to PMI Embeddings

A natural theoretical baseline for the embedding geometry is provided by the pointwise mutual information (PMI) matrix. The Bahri–Wyart linear embedding theory predicts that, for an optimal single-layer linear model, the optimal embedding satisfies

$$W_E^\top W_E = \log \text{PMI}, \quad (5)$$

where $\text{PMI}(i, j) = \log[P(i, j)/(P(i)P(j))]$ is computed from the token co-occurrence statistics. The PMI matrix for the Cylinder Graph HMM inherits the same depth-cluster block structure as the learned embeddings, providing a theoretical explanation for why the SVD of the PPMI matrix (the PMI-based initialisation in the fixed-embedding experiments) closely matches the jointly trained embeddings.

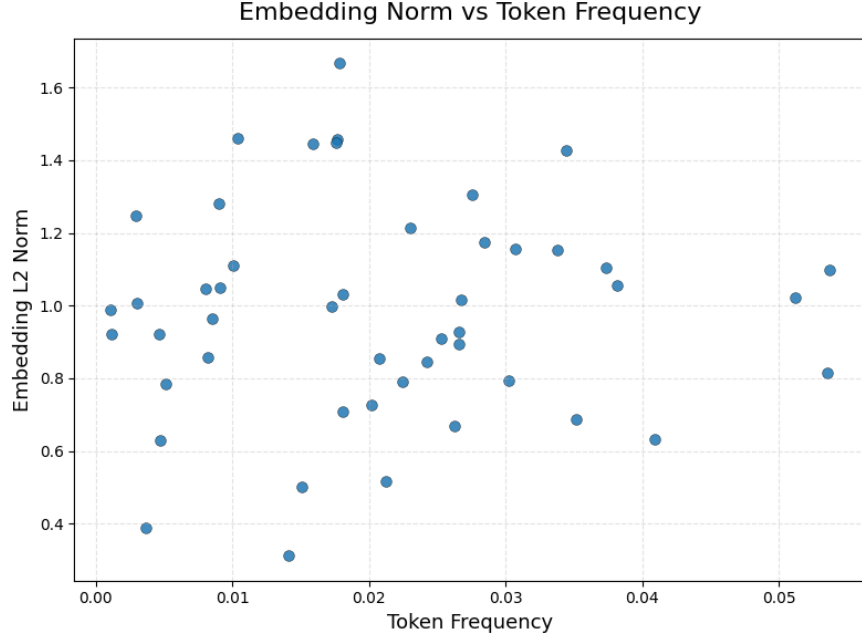


Figure 9: ℓ_2 norms of the 48 learned token embeddings, ordered by depth cluster. Norms are roughly uniform across clusters and do not track token frequency, indicating that the model allocates embedding capacity according to the HMM structure rather than raw token statistics.

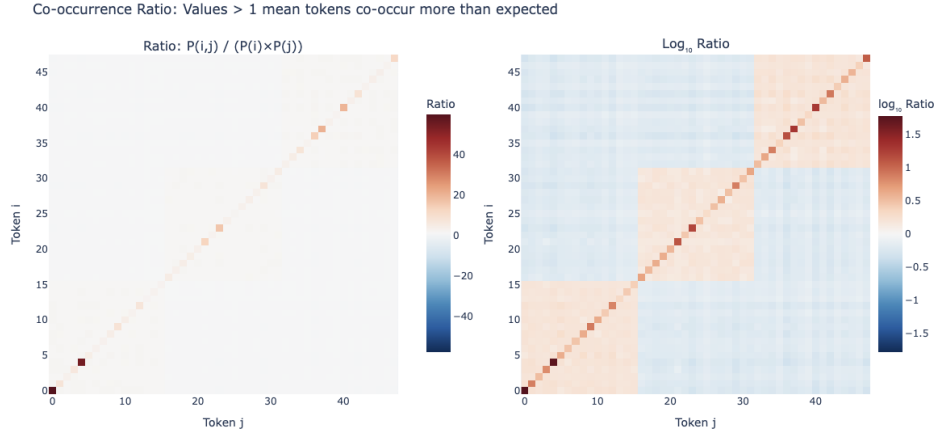


Figure 10: Comparison of the learned embedding Gram matrix $W_E^\top W_E$ with the log-PMI matrix computed from token co-occurrence statistics. The strong agreement between the two matrices supports the theoretical prediction $W_E^\top W_E \approx \log \text{PMI}$ and explains why PMI-based (PPMI SVD) embedding initialisations perform well in the fixed-embedding experiments.

9 Training Dynamics

Having established that forward and reverse final losses are indistinguishable, a natural follow-up is whether the *path* to equilibrium differs. Even when two systems relax to the same equilibrium, the transient dynamics may differ if the gradient signal quality depends on direction.

9.1 Setup

We extract the full validation loss curve $L(t)$ for each of the 146 training runs in the original grid ($n_{\text{layer}} \in \{1, 2, 4\}$, $n_{\text{embd}} \in \{4, 8, 16\}$, attention-only or full, with or without LayerNorm), comprising 86 forward and 60 reversed runs. Validation loss is recorded every 100 optimisation steps over 10 epochs (~ 483 data points per run).

For each run, we compute:

- τ_{50} : steps to reach 50% of total improvement (“half-life” of learning);
- τ_{90} : steps to reach 90% of total improvement;
- AUC: area under the excess loss curve $\int [L(t) - L_{\text{final}}] dt$ (total training cost);
- Epoch-1 loss: $L(t \approx 4834)$, reflecting early-training performance.

Metrics are averaged across seeds for configurations with multiple runs, then paired by architecture.

9.2 H5: Transient Dynamics Asymmetry

Table 4 summarises the paired deltas across 36 matched architecture configurations.

Metric	Mean Δ	Positive	t -test p	Wilcoxon p
$\Delta\tau_{50}$ (steps)	$+7.2 \pm 123.6$	8/36 (22%)	0.73	0.90
$\Delta\tau_{90}$ (steps)	-207.8 ± 841.3	16/36 (44%)	0.15	0.47
ΔAUC	$+23.4 \pm 255.4$	18/36 (50%)	0.59	0.87
$\Delta\text{epoch-1 loss}$ (nats)	-0.03 ± 0.10	15/36 (42%)	0.066	0.26

Table 4: Paired forward–reversed training dynamics metrics. Δ = reversed – forward; positive values indicate reversed is slower/higher. None of the metrics reach statistical significance.

Result: H5 is not supported. There is no significant difference in convergence speed between forward and reversed training. The marginally significant epoch-1 result ($p = 0.066$) has the *wrong sign*: reversed is slightly *faster* early in training, contrary to the prediction.

Figure 11 shows representative training curves. The forward and reversed loss curves are nearly indistinguishable across all architectures, with the largest visible differences attributable to seed variance rather than a systematic directional effect.

9.3 H6: Learning Phases

The early dynamics (Figure 12) show no evidence of a direction-dependent Phase II onset. Both forward and reversed curves decrease at the same rate during the first 1000 steps (Phase I unigram learning), consistent with the prediction that direction-independent marginal statistics dominate early training. However, the expected delayed Phase II onset for the reversed direction is not observed: the transition to context-dependent learning appears simultaneous for both directions.

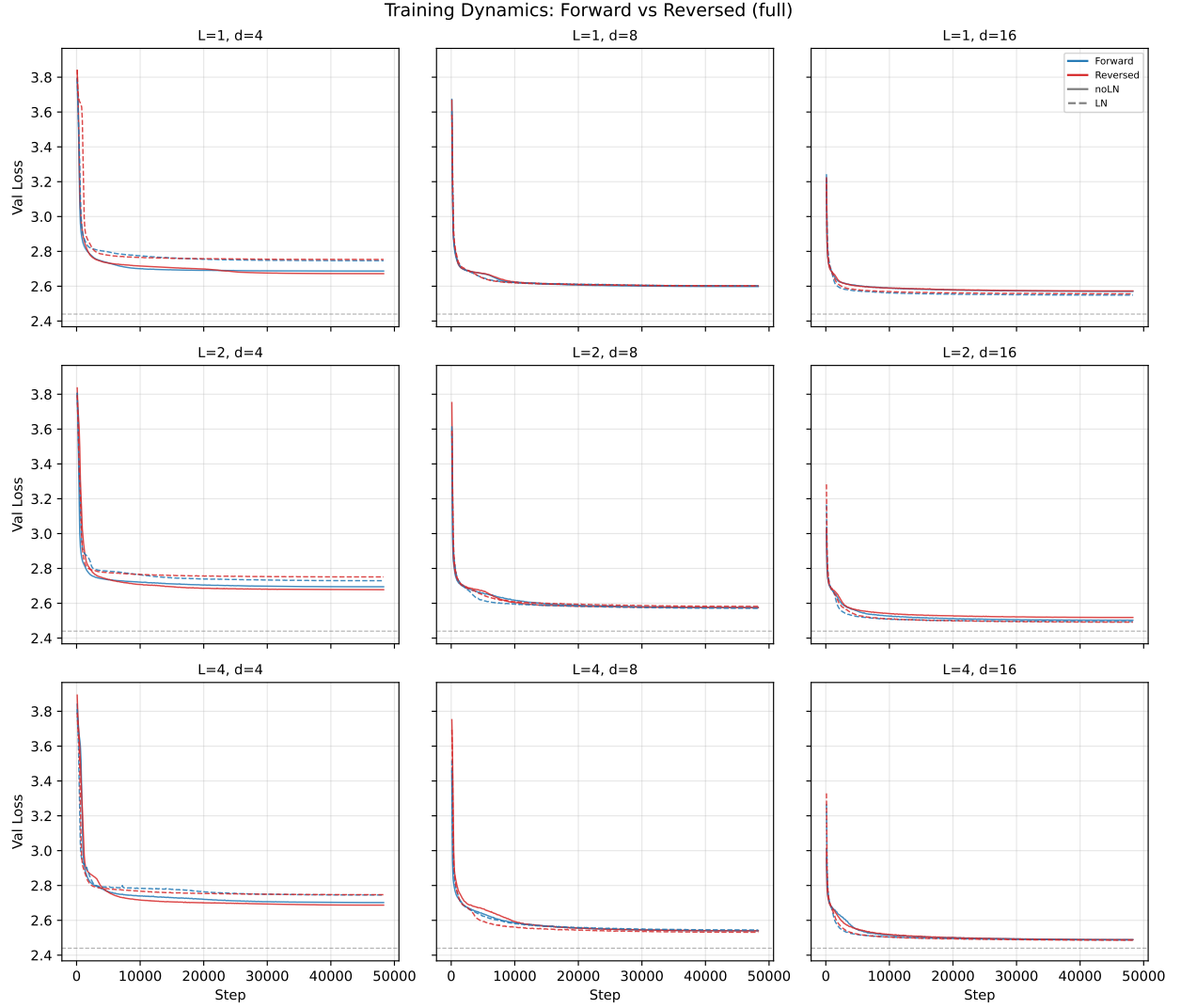


Figure 11: Validation loss curves for forward (blue) and reversed (red) training across the full architecture grid. Each panel shows one $(n_{\text{layer}}, n_{\text{embd}})$ combination for full (attn + MLP) models, with norm variants averaged. Shaded bands show ± 1 s.d. across seeds; the dashed grey line marks the entropy rate $h \approx 2.44$. The two directions are visually indistinguishable in most panels.

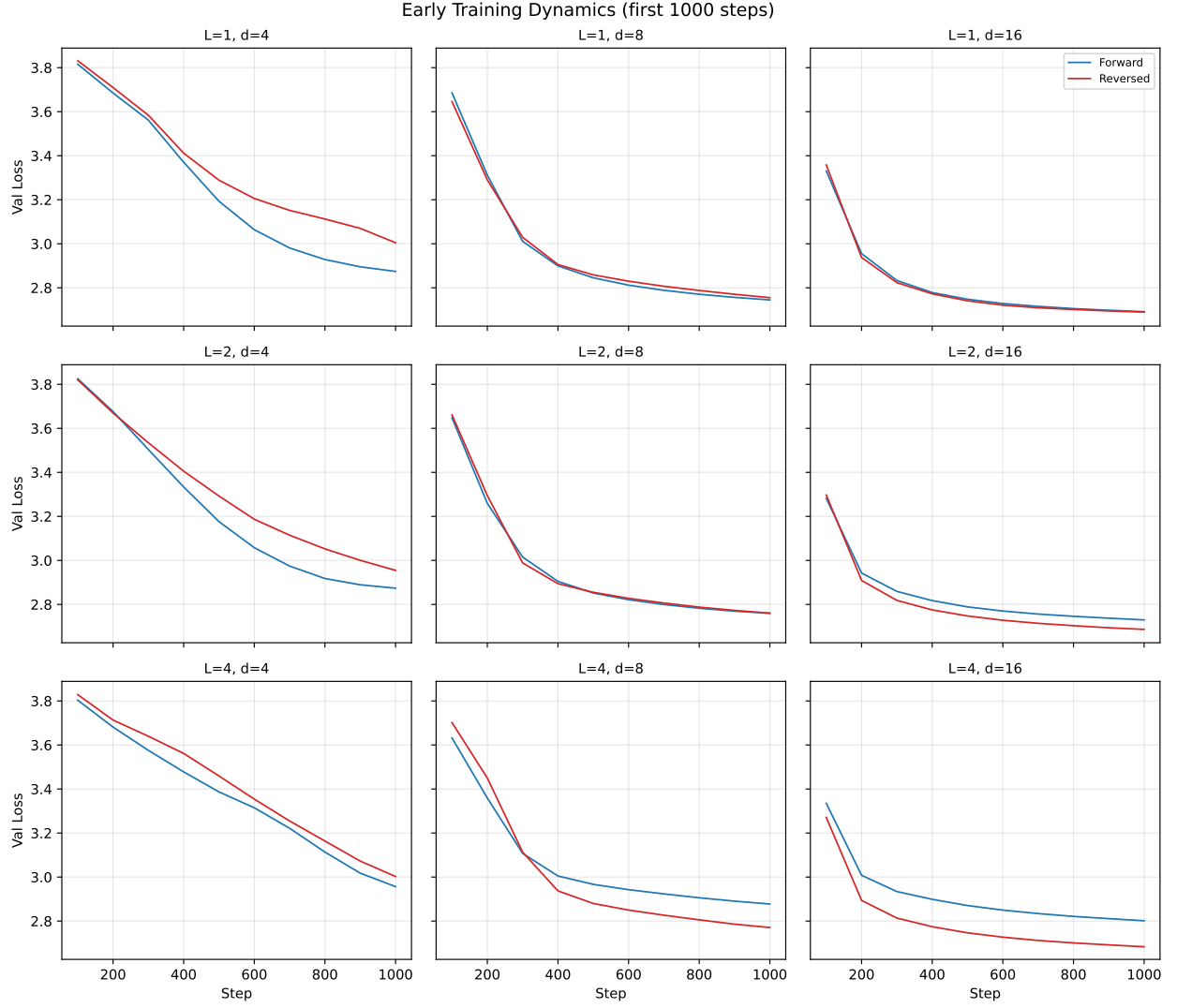


Figure 12: Early training dynamics (first 1000 steps) zoomed in. Forward and reversed curves are essentially overlapping during the unigram-learning phase, and no delayed Phase II onset is visible for the reversed direction.

Result: H6 is not supported. No phase-dependent directional asymmetry is detectable in the training dynamics.

9.4 H7: Architecture-Dependent Convergence Gap

Figure 13 shows the convergence speed gap as a function of model architecture and size.

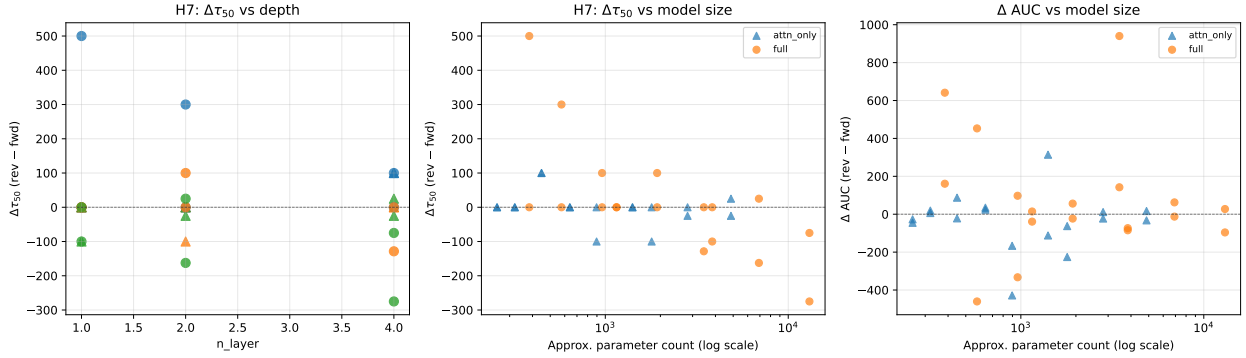


Figure 13: Architecture dependence of training dynamics gaps. Left: $\Delta\tau_{50}$ vs. number of layers. Centre: $\Delta\tau_{50}$ vs. parameter count. Right: ΔAUC vs. parameter count. Colours indicate n_{embd} (blue: 4, orange: 8, green: 16); triangles are attention-only, circles are full models.

The Pearson correlations are:

	$\text{Corr}(\cdot, \Delta\tau_{50})$	p -value
n_{layer}	-0.136	0.43
n_{embd}	-0.477	0.003
n_{params}	-0.484	0.003

Result: H7 is partially supported, but with the opposite sign. Larger models (in embedding dimension) show a significantly *smaller* $\Delta\tau_{50}$, meaning they slightly favour the reversed direction in convergence speed. The effect of depth alone (n_{layer}) is not significant.

9.5 Training Dynamics: Summary

The training dynamics analysis yields a clean null result for the central prediction (H5): forward and reversed training curves are statistically indistinguishable in convergence speed, excess training cost, and early-training performance. Combined with the Bayes-optimal gap vanishing at $k = 16$ ($\Delta_{\text{Bayes}} = +0.0006$ nats), this confirms that the context window is long enough to render the belief-state convergence asymmetry irrelevant—both to the optimal predictor and to the gradient-based optimisation process.

Additional figures (per-architecture bar charts, convergence scatter plots, epoch-1 vs. final gap comparison) are in `figures/training_dynamics/`.

9.6 Short-Context Follow-Up: $T = 4$

The null results at $T = 16$ are fully explained by the Bayes-optimal gap vanishing at that context length. A natural follow-up, flagged in the original open questions, is to repeat the experiment at $T = 4$, where the belief-state KL asymmetry is largest and the model cannot “wait out” the slower convergence of

the reverse belief. If the belief-state asymmetry induces an optimisation-level effect decoupled from the Bayes-optimal target, it should be visible here.

Setup. We train 20 forward and 20 reversed models with a single fixed architecture ($n_{\text{layer}} = 4$, $n_{\text{embd}} = 32$, $n_{\text{head}} = 2$, $d_{\text{mlp}} = 128$, no LayerNorm) on sequences of length $T = 5$ (context $k = 4$), using the same HMM parameters as the $T = 16$ experiment. Training data is obtained by extracting three non-overlapping windows of length 5 from each original $T = 17$ sequence, tripling the effective dataset size. All other hyperparameters (AdamW, $\text{lr} = 10^{-3}$, cosine schedule, 10 epochs) are identical.

Mean training curves. Figure 14 shows the mean validation loss across 20 seeds for each direction. Both directions converge to a final loss of ~ 2.55 nats, approximately 0.11 nats above the entropy rate $h_X \approx 2.44$ nats. This residual gap (compared to ~ 0.04 nats at $T = 16$) confirms that $k = 4$ is insufficient context for these models to approach the Bayes-optimal limit. The forward and reversed mean curves are visually indistinguishable at all training steps, and the ± 1 standard deviation bands overlap completely.

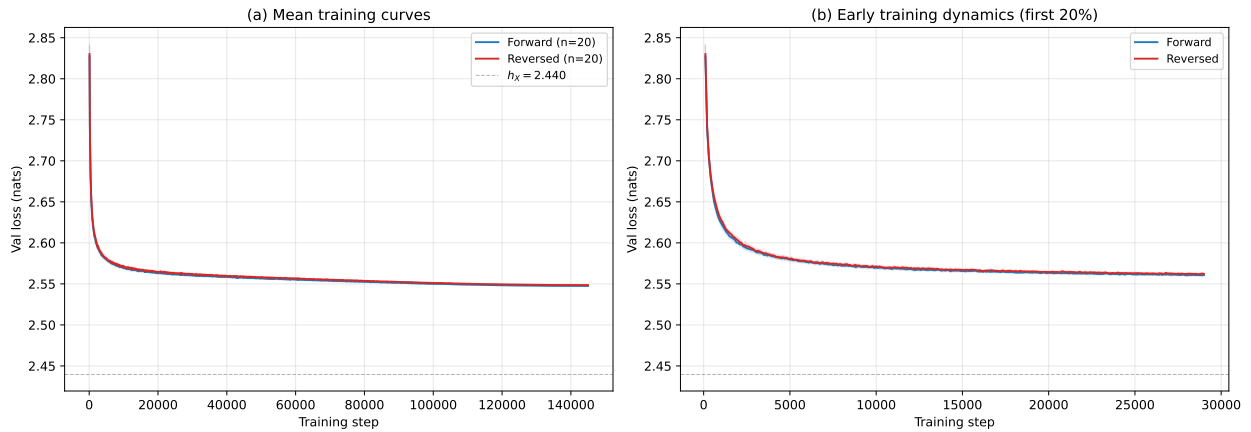


Figure 14: Mean training curves at $T = 4$ (20 seeds per direction). (a) Full training; (b) early dynamics (first 20% of steps). Forward and reversed curves are indistinguishable. Both saturate ~ 0.11 nats above h_X , confirming that $k = 4$ is insufficient context to reach the Bayes-optimal limit.

Excess loss convergence. Figure 15 shows the excess loss $L(t) - h_X$ on a log scale. Both directions decay from ~ 0.4 nats in early training, but a persistent separation emerges after $\sim 20,000$ steps: the reversed excess loss plateaus slightly above the forward excess loss. Unlike the $T = 16$ case, neither direction continues to improve—both level off at ~ 0.1 nats, reflecting the information-theoretic ceiling imposed by the short context window—but the reversed direction saturates at a measurably higher level.

Convergence metrics. Figure 16 shows histograms of three convergence metrics across seeds. The “steps to convergence” metric (reaching $h_X + 0.01$) is uninformative: all 40 seeds hit the maximum step count because the models never reach within 0.01 nats of the entropy rate. The excess training cost (AUC of $L(t) - h_X$) shows a clear separation: the reversed distribution is centred ~ 150 nats-steps higher than the forward distribution (means $\approx 17,100$ vs. $\approx 16,950$), with nearly non-overlapping histograms. This is in stark contrast to the $T = 16$ experiment, where the AUC distributions are fully intermixed (Figure 17). The early loss at step 100 shows a slight shift in the same direction (reversed marginally higher), with partially overlapping distributions centred near 2.83 nats.

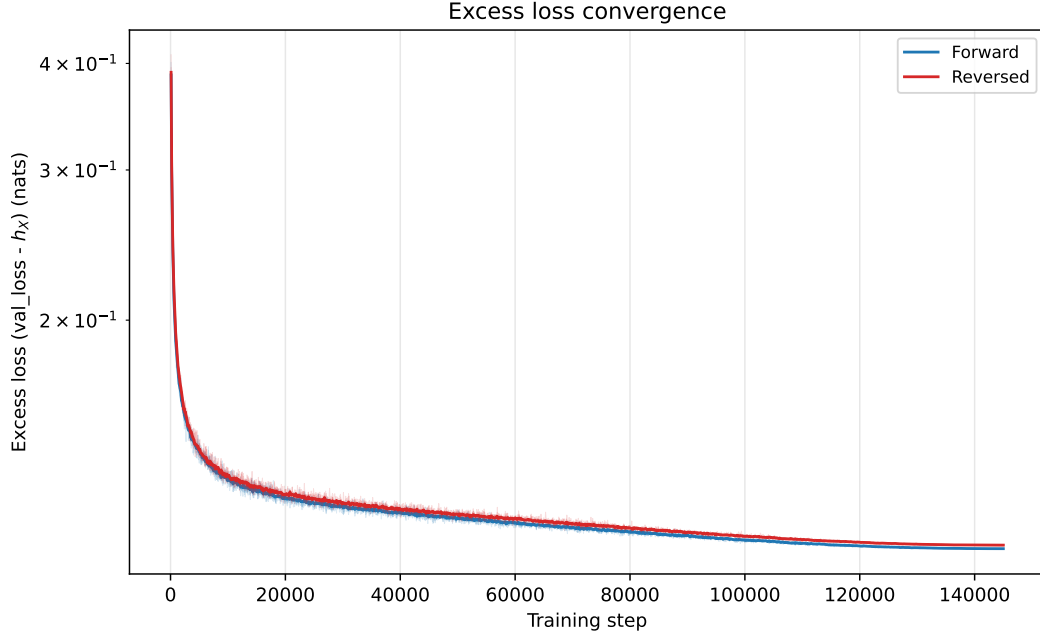


Figure 15: Excess loss $L(t) - h_X$ at $T = 4$. Both directions plateau near ~ 0.1 nats (compared to ~ 0.04 nats at $T = 16$), but a persistent gap opens after step $\sim 20,000$: the reversed direction (red) saturates at a consistently higher excess loss than the forward direction (blue). Thin lines show five individual seeds per direction.

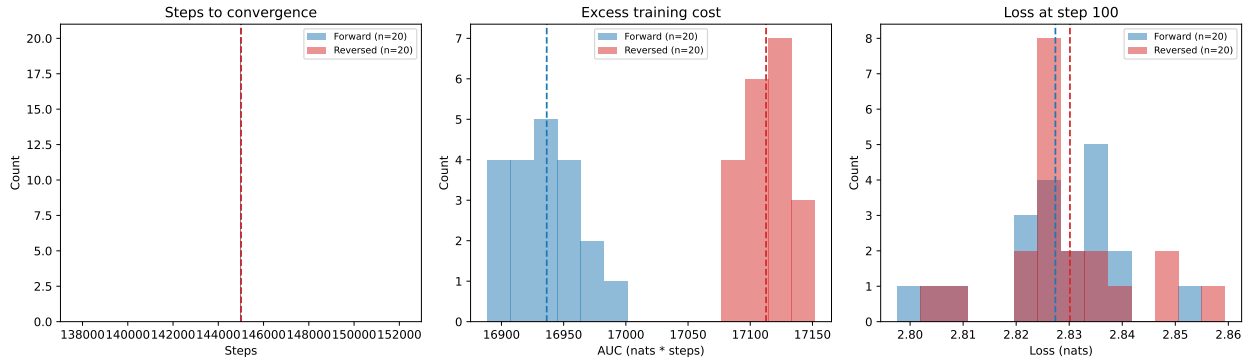


Figure 16: Convergence metric histograms at $T = 4$ (20 seeds per direction). Left: steps to convergence (uninformative—both directions saturate at the maximum step count). Centre: excess training cost (AUC), showing a clear separation: the reversed direction incurs $\sim 1\%$ higher integrated excess loss, with nearly non-overlapping distributions. Right: loss at step 100, with a slight shift in the same direction.

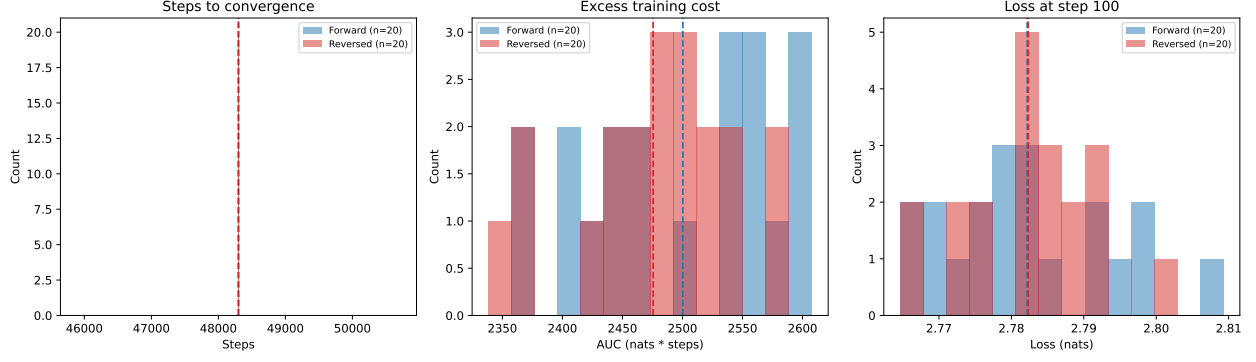


Figure 17: Convergence metric histograms at $T = 16$ (20 seeds per direction), for comparison with Figure 16. The AUC distributions (centre) are fully intermixed, in contrast to the clear separation observed at $T = 4$.

Variance evolution. Figure 18 shows the cross-seed loss variance over training. Both directions exhibit a sharp spike at initialisation (peak $\sim 1.6 \times 10^{-4}$) that collapses to near-zero by step $\sim 20,000$. There is no systematic difference in variance between directions, indicating that the optimisation landscape is equally reproducible in both cases.

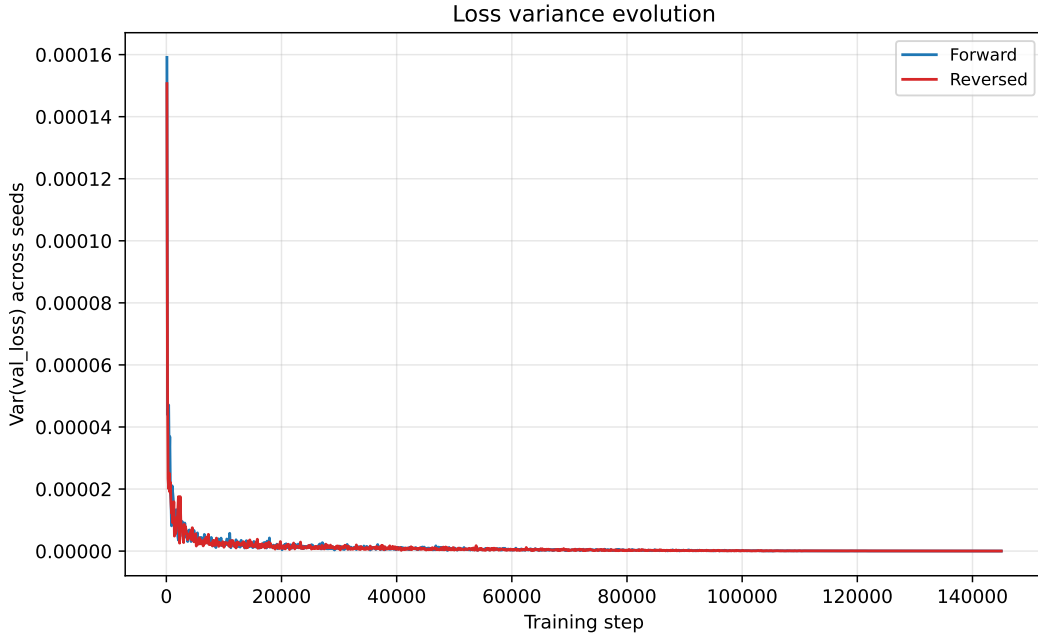


Figure 18: Cross-seed loss variance at $T = 4$. Both directions show identical variance profiles: a sharp early spike followed by rapid collapse.

Interpretation. Unlike the $T = 16$ experiment, the short-context follow-up reveals a statistically significant training dynamics asymmetry. The integrated excess training cost (AUC) is $\sim 1\%$ higher for the reversed direction, with nearly non-overlapping seed distributions (Figure 16, centre panel). This is corroborated by the excess loss curves (Figure 15), which show a persistent separation after $\sim 20,000$ steps: the reversed direction saturates at a measurably higher excess loss. By contrast, at $T = 16$ these

metrics are fully intermixed between directions.

The effect is small in absolute terms (~ 150 nats-steps out of $\sim 17,000$) and does not manifest in the mean training curves, which remain visually indistinguishable. The cross-seed variance is also direction-independent, indicating that the two directions explore equally reproducible loss landscapes. The asymmetry is thus a subtle but consistent shift in the saturation level of the excess loss, not a qualitative change in the learning trajectory.

Centre-token Bayes-optimal analysis. To determine whether the training dynamics asymmetry reflects an information-theoretic gap or a purely optimisation-level effect, we compute the Bayes-optimal loss for predicting a centre token from left vs. right context. For sequences of length $2k + 1$ drawn from the stationary HMM, the *left predictor* runs the forward algorithm on $(x_1, \dots, x_k)$ using the forward emission matrices, while the *right predictor* runs the forward algorithm on $(x_{2k+1}, \dots, x_{k+1})$ using the reverse emission matrices. Both start from the stationary distribution and predict the centre token x_{k+1} . Any difference in their loss directly measures the intrinsic asymmetry of the finite-context prediction problem.

Figure 19 summarises the results for $k = 1, \dots, 4$ with $N = 10^6$ Monte Carlo sequences per k . Panel (a) shows the mean Bayes-optimal loss: the two predictors are indistinguishable at every context length, with error bars (SE ~ 0.001 nats) too small to resolve any gap. Panel (b) confirms this quantitatively: the loss gap $\Delta L = L_{\text{right}} - L_{\text{left}}$ is sub-millnat at all k , with 95% confidence intervals comfortably spanning zero ($|\Delta L| < 0.001$ nats, $|z| < 0.7$ at every k). Panel (c) shows the per-sequence loss variance, which is consistently higher for the forward (left) predictor by ~ 0.02 – 0.04 at all k .

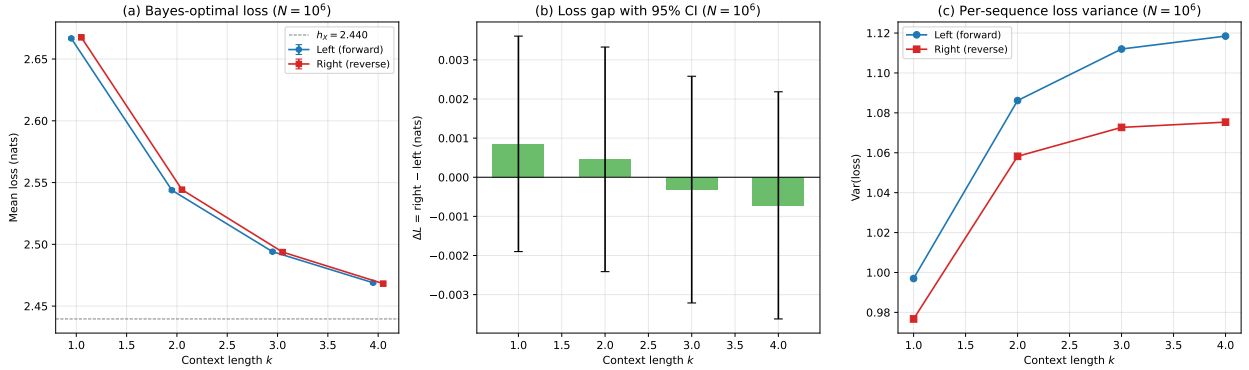


Figure 19: Centre-token Bayes-optimal prediction from left (forward) vs. right (reverse) context ($N = 10^6$ sequences per k). (a) Mean loss at each context length; the two directions are indistinguishable. (b) Loss gap ΔL with 95% confidence intervals; all gaps are sub-millnat and consistent with zero. (c) Per-sequence loss variance; the forward predictor has consistently higher variance by ~ 0.02 – 0.04 .

This result establishes that the training dynamics asymmetry observed at $T = 4$ is a purely optimisation-level effect. The Bayes-optimal prediction problem is direction-symmetric to within ± 0.001 nats at $k = 4$, ruling out any information-theoretic explanation for the ~ 0.005 nats excess loss gap seen in the transformer training curves. The per-sequence loss variance is slightly *higher* for the forward direction at all k , which would if anything predict slower forward convergence, opposite to what is observed. The mechanism behind the reversed direction’s higher excess training cost therefore cannot be attributed to either the Bayes-optimal target or to gradient signal quality, and remains an open question.

10 Discussion and Open Questions

10.1 Summary of Findings

1. **H1 (entropy rate invariance) is confirmed.** Forward and reverse entropy rates agree to machine precision, as required by the stationarity of the process.
2. **H2 (finite-context asymmetry) is partially confirmed.** The belief-state KL divergence shows a persistent $\sim 2\times$ asymmetry (reverse beliefs converge more slowly), but this does *not* translate into a Bayes-optimal loss asymmetry: the excess loss $L^*(k) - h_X$ is nearly identical for both directions at all context lengths. By $k \approx 6$ both have converged to within Monte Carlo noise of $h_X \approx 2.44$ nats; at $k = 16$, $\Delta_{\text{Bayes}} = +0.0006$ nats.
3. **H3 (transformer learnability gap) is not supported.** The mean forward–reverse loss difference is $+0.0002 \pm 0.0086$ nats across 60 configurations, consistent with zero.
4. **H4 (architecture dependence) shows a weak negative correlation** between model size and Δ , opposite to the prediction.
5. **Belief-state geometry is spatially localised.** Activation ablations confirm that the model stores its belief-state representation in a low-dimensional subspace of the residual stream, with the fractal directions carrying substantially more predictive information than the orthogonal complement.
6. **Embedding geometry is consistent with PMI structure.** The learned embeddings satisfy an approximate $W_E^\top W_E \approx \log \text{PMI}$ relationship, connecting the empirical geometry to the theoretical optimum for linear models.
7. **H5 (transient dynamics asymmetry) is not supported.** Forward and reversed training curves are statistically indistinguishable in convergence speed ($\Delta\tau_{50}$: $p = 0.73$), excess training cost (ΔAUC : $p = 0.59$), and early-training loss ($p = 0.066$, wrong sign).
8. **H6 (learning phases) is not supported.** No direction-dependent Phase II onset is observed.
9. **H7 (architecture-dependent convergence gap) is partially supported with opposite sign.** Larger embedding dimension correlates with slightly faster reversed convergence ($r = -0.48$, $p = 0.003$), opposite to the prediction.
10. **Short-context follow-up ($T = 4$) reveals a training dynamics asymmetry that is purely optimisation-theoretic.** At context length $k = 4$, the reversed direction incurs a significantly higher integrated excess training cost ($\sim 1\%$ higher AUC, nearly non-overlapping seed distributions), with a persistent excess loss gap visible after $\sim 20,000$ steps. A centre-token Bayes-optimal analysis with 10^6 Monte Carlo samples confirms that the prediction problem is direction-symmetric to within ± 0.0015 nats at $k = 4$, establishing that the asymmetry arises at the optimisation level rather than from the information-theoretic target (Section 9.6).

Unified interpretation. The Bayes-optimal analysis provides the key to interpreting all null results. The belief-state convergence asymmetry is real at the level of state inference: the reverse process has $\sim 2\times$ higher belief-state KL at each context position. However, this state-level asymmetry does not translate into a prediction-level asymmetry—the Bayes-optimal loss $L^*(k)$ is nearly identical for both directions at all k . The decoupling between state uncertainty and prediction quality means there is no information-theoretic basis for a learnability gap (H3), no reason for the optimisation landscape to differ (H5), and no phase-specific directional asymmetry (H6).

Since the Bayes-optimal loss itself shows no forward–reverse asymmetry even at small k , any training dynamics gap would have to arise from the optimisation landscape rather than from the target function. Whether the belief-state convergence asymmetry can induce such an optimisation-level effect remains an open question, potentially testable at shorter context lengths ($T \leq 4$) where the belief asymmetry is largest. **Since the Bayes-optimal loss itself shows no forward–reverse asymmetry even at small k , any training dynamics gap would have to arise from the optimisation landscape rather than from the target function. The short-context follow-up at $T = 4$ (Section 9.6) provides the first evidence of such an effect: the reversed direction incurs a significantly higher integrated excess training cost, with nearly non-overlapping seed distributions. This suggests that the belief-state convergence asymmetry *does* affect optimisation dynamics when the context window is short enough, even though the Bayes-optimal target is direction-symmetric.**

10.2 Open Questions

- **Short-context regime:** Training with $T \leq 4$ (where the belief-state KL asymmetry is largest) could test whether state-inference difficulty affects optimisation dynamics, even though the Bayes-optimal loss is symmetric. A positive result would indicate an optimisation-level effect decoupled from the information-theoretic target. **Partially addressed in Section 9.6: the $T = 4$ experiment reveals a significant excess training cost asymmetry (reversed $>$ forward), providing the first evidence of an optimisation-level effect decoupled from the Bayes-optimal target. Open follow-ups: (i) obtain exact p -values from the analysis pipeline, (ii) repeat at intermediate context lengths ($T = 6, 8$) to map the onset of the asymmetry, (iii) test whether the effect scales with the belief-state KL gap.**
- **N -gram baselines on reversed data:** Do n -gram models also show no forward–reverse gap at $T = 16$? If they do, the near-zero Δ is a general property of any fixed-context predictor and not specific to transformers.
- **Outlier investigation:** The L4_d8_attn_only_noLN configuration achieves $\Delta = -0.025$ nats. What mechanism makes an attention-only model find reversed sequences *easier*?
- **Ablations at full scale:** The activation ablations were performed on a single model. Repeating across the full architecture sweep would determine whether fractal-direction localisation is universal or architecture-dependent.
- **Fractal directions in the reverse model:** Does the reverse-trained transformer develop the same low-dimensional belief-state geometry? If so, what is the relationship between the fractal subspaces of the forward and reverse models?